

MD. ASHIKUR RAHMAN BISHAL

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Frontend-leaning software Engineer with **3+** years of experience, collectively serving **85M+** end users across **190+** financial institutions. Focused on growth, and fostering collaborative learning environments.

INDUSTRY EXPERIENCE

DYNAMIC SOLUTION INNOVATORS LTD.

April, 2023—Present

Software Development Unit

- Accelerated development workflows by integrating **AI-assisted tooling**, cutting repetitive overhead and shipping features faster across multiple projects.
- Mentored junior engineers through structured **technical coaching**, including hands-on code reviews, pair programming, and knowledge-sharing sessions, **improving overall team capability**.
- Performed **in-depth security auditing** and applied mitigation strategies to **resolve application-level vulnerabilities**.
- Proactively built **helper tools** adopted across the team to automate repetitive workflows; increasing efficiency, and reducing delivery time.
- Identified and resolved critical **performance bottlenecks** across multiple applications, optimizing **API call efficiency**, reducing unnecessary network calls.
- Took **ownership of critical production issues**, resolving incidents reducing downtime through effective communication and coordination.
- Demonstrated strong **accountability and data ethics** in handling sensitive, and bulk banking data; strictly adhering to **data privacy standards** and regulatory compliance requirements.
- **Led migration** of legacy monolithic architecture to a modern multi-tenant platform, delivering a fully tested and documented transition across multiple services with zero critical disruptions.
- Collaborated with **cross-functional** and distributed teams across multiple projects to deliver reliable, scalable, and maintainable software solutions.
- Consistently produced **clean, well-structured code** and comprehensive documentation, upholding high engineering standards to support long-term maintainability and team knowledge sharing.

RESEARCH INVOLVEMENTS

MINISTRY OF ICT DIVISION

January, 2022—October, 2022

Research Internship

- Formulated the **mathematical scoring model** for a financial reputation engine, deriving logarithmic scoring functions over money transfer, loan repayment, and bill payment behaviour.
- Designed the **aggregation function** combining these scores with account age, bounded to a normalized range using a modified sigmoid transformation.
- Implemented the engine as **recursive time-indexed algorithms** with thorough documentation.

UNDERGRADUATE THESIS

March, 2022—March, 2023

BlockURS: A Blockchain-based System for a Unified Reputation Score

- Performed a **systematic survey and critical analysis** of state-of-the-art reputation systems.
- Conducted **threat modeling using the STRIDE framework**, deriving formal functional, security, and privacy requirements for the proposed system.
- Formalized **mathematical reputation scoring models** tailored to specific domains, eventually unifying under a single cross-domain aggregation function.
- Architected a private **blockchain network with smart contracts**, to ensure decentralized, tamper-resistant management of reputation data.
- Designed **secure protocol flows**, including registration, authentication, reputation retrieval, and data provisioning, using **public-key cryptography** and nonce-based replay attack mitigation techniques.

PUBLICATIONS

UNDER REVIEW

[1] **A Blockchain Empowered & Reputation Integrated e-KYC System** 2026

Blockchain: Research and Applications (Elsevier) · Manuscript Number: BCRA-D-26-00874

Md Yeasin Ali, Abrar Fahim, Romana Mahjabin Eshita, Kiriti Mukherjee, **Md. Ashikur Rahman Bishal**, Md. Masum Alam Nahid, Mohammad Javed Morshed Chowdhury, Mohammad Shahriar Rahman, and Md Sadek Ferdous

COMPETENCES

Programming	JavaScript, TypeScript, C++
Web Stack	React.js, Next.js, Vue.js, Meteor.js, HTML5, CSS3, TailwindCSS
Databases	Oracle, PostgreSQL
Tools	Git, SVN, Postman, JIRA, Datadog, Figma
AI Workflow	Claude Code, Antigravity, Codex

NOTABLE PROJECTS

FRONTEND 101

2026—Present

 [Github Repository](#)  [Live Demo: frontend101.arbishal.com](#)

Creator & Sole Maintainer

- Initiated and built an **interactive simulation platform** teaching frontend fundamentals through live, manipulable demonstrations paired with written explanations.
- Designed the **concept sequencing and explanatory material** across the curriculum, structuring topics from first principles to composition.

TypeScript

Next.js

TailwindCSS

Shiki

Claude Code

FIREFLIES SIMULATION

2025

 [Github Repository](#)  [Live Demo: fireflies.arbishal.com](#)

Passion Project

- Developed an **interactive firefly simulation**, preserving their magic through generative motion and flickering light.
- Implemented **touch/cursor reactive physics**—fireflies drift, pulse, and gently attract or scatter based on interaction; with real-time controls for population, radius, speed, etc.

JavaScript

CanvasAPI

Vue.js

TailwindCSS

EDUCATION

B. Sc in Engineering

Shahjalal University of Science & Technology

2023

CGPA: **3.67/4.00**

Major: *Computer Science & Engineering*

INTERESTS

Research Focus

Blockchain & Distributed Systems, Practical AI Applications for Productivity

Creative Pursuits

Storytelling & Creative Writing, Graphic Design, Photography